

encryption or Bob's decryption program works, she's back in the game—all she has to do is reverse-engineer their methods.

So plain old gibberish-ing won't work for Alice and Bob—they need a *key* to lock their message. The key is basically a string of numbers that the encryption program uses when it's turning messages into ciphertext. All mathematical operations rest on this key, and so even though Eve can still access the ciphertext *and* know exactly how the encryption and decryption programs work, she is still baffled without it.

Protect The Key, You Must

Even though Alice and Bob have keys that they can now mockingly wave in Eve's face, the big question for Alice is, "If I write my message using a key, how do I send the key to Bob so he can read it?" Obviously, sharing the key should be done in complete secret, but they really can't plan for Eve's creativity when it comes to new and improved methods to eavesdrop.

The first idea is the Public key and Private Key system. This way, Alice and Bob don't need to hunt for secure ways to distribute their keys. Instead, they each have a *Private Key* which they use to decrypt messages, and a *Public Key* which is used to encrypt the message. Think of it as a box with two keys—anyone with the public key can lock it, but only the person with the private key can unlock it. The public key/private key system relies on the fact that there are some mathematical operations that take longer to undo than do—for example, you could go all calc-happy and keep multiplying numbers to get a 1000-digit number, but factorising that number into all the numbers you started with is a pain even for supercomputers! Say Alice wants to send a message to Bob:

Alice: Oi! I need to send you something. Give me your public key.

Bob: Sure; it's 12I94732

Eve [with evil grin]: Hee hee, now I know the key!

Alice will then use Bob's public key to encrypt her message and send it to Bob.

Alice: "S*&?"{|\?0%&

Bob: Aha! Now that the message has been locked with *my* public key, I can use my own *private* key to unlock it.

Today's computers would take ages to crack an encrypted message, but what about tomorrow's computers?

Eve: What the...? I can't decrypt this!

Another method is the terribly secret *One-Time Pad* (OTP). The OTP is a random key, the same length as the message, which is applied once, and only once. Theoretically, there is no way of deciphering an OTP-encrypted message without knowing the pad itself. There is, of course, the fact that this pad has to be guarded with great care. Out of sheer paranoia, this pad also needs to be: (a) completely random so that Eve can't detect patterns and figure out the pad herself, and (b) used only once, so it's useless even if Eve finds it out—keeping her safely out of the conversation.

Sitting Pretty?

It all sounds really safe, doesn't it? To be dead honest, cryptography as it exists today is quite sufficient for our causes. Breaking into current systems is quite tough; that people have enough faith to carry out billions of dollars worth of transactions over the Internet speaks for itself.

But safety now doesn't mean safety forever. True, today's computers would take ages to crack an encrypted message (see box *Attack!*), but what about tomorrow's computers? Quantum computing is going to turn complex mathematics into child's play. Think of the 1000-digit number problem—a quantum computer would actually be able to give *all* the factors to the number in a single shot!

The first to succumb will be the public key/private key system. Nobody can deny that there is a relationship between the public and private keys of a person—after all, it is because of this relationship that enables people to tell whether a message is meant for them. It still hasn't been proven that Eve can't 'pick' Alice's lock just by knowing her public key and using math to figure out the private key. As computers become more powerful, scientists will have to keep refining this technique to keep private keys private.

The One-time pad is quite secure—in fact, it has been proven that even if Eve had infinite computing power, she wouldn't be able to break it. The trouble is, the key needs to be shared in a way that is *perfectly secure*, otherwise the entire effort is pointless.

Everything else aside, there is still no way to tell if your communication line is being monitored or not. So apart from guarding against the computers of the future, we need to figure out a way to prevent eavesdroppers getting out ciphertext at all.

Thankfully, before quantum computing arrives and exposes all our embarrassing secrets, there will be *Quantum Cryptography*, which will ensure that the Alices and Bobs of the future can still chat without Eve ruining the party.

Light In Shining Armour

Rather than using mathematics to encrypt messages, scientists have been working on quantum cryptography, which uses the quantum-physical properties of light to aid in securing communications. The purpose of quantum cryptography is to provide a secure way to transfer keys between Alice and Bob without Eve being able to find them out.

Attack!

There are many methods that dedicated hackers can use to eavesdrop on your information.

The *Brute-Force Attack* is the most basic of them all. It runs through all possible combinations of numbers and letters to figure out the key to decrypt messages. Naturally, it's really inefficient and can actually take thousands of years to crack today's codes.

In a *Man-in-the-middle Attack*, the hacker intercepts messages between both parties, maybe even sending out wrong information to them without being detected. For example, in the public/private key system, Eve could intercept a message from Alice to Bob which contains Alice's public key. She could then send Bob her own public key. Bob would then use *Eve's* public key to send his secret message to who he thinks is Alice, but in reality, he's making it for Eve! Eve could then use Alice's public key to encrypt anything she wants Bob to read. Through all this, Alice and Bob don't know that Eve is listening in, so they think that they are talking to each other.